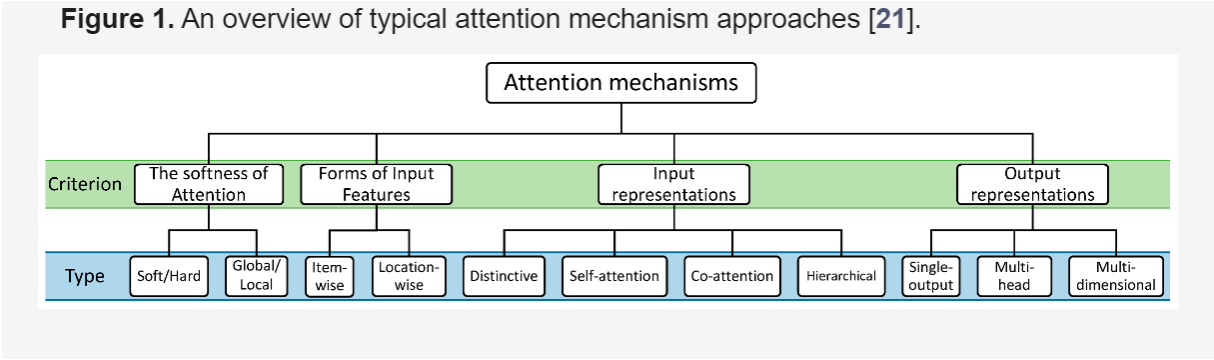


In the domain of sequence-to-sequence models, traditional architectures relying on recurrent or convolutional layers face challenges in encoding lengthy sequences and preserving context effectively. The integration of attention mechanisms addresses these limitations, offering a groundbreaking solution. This not only mitigates the challenges but also provides models with the capability to dynamically focus on essential parts of the input sequence.

Figure 1. An overview of typical attention mechanism approaches [21].



The attention mechanism empowers models to discern and assign varying degrees of importance or "attention" to distinct parts of the input sequence during the prediction process. Instead of treating all elements equally, attention allows the model to selectively focus on the most pertinent information.